Zhiran Qin

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Dept. of Agricultural and Resource Economics
PhD-27
University of California, Davis
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Education

Ph.D. in Agricultural and Resource Economics, UC Davis — (Anticipated) June 2027

B.A. in Agricultural Economy and Management, Renmin University of China — June 2021

Dissertation Committee

Prof. Pierre Mérel (**Co-chair**)
Department of Agricultural and Resource Economics
University of California, Davis
prmerel@ucdavis.edu

Distinguished Prof. Richard J. Sexton (**Co-chair**)
Department of Agricultural and Resource Economics
University of California, Davis
rjsexton@ucdavis.edu

Assistant Prof. James E. Sayre
Department of Agricultural and Resource Economics
University of California, Davis
jsayre@ucdavis.edu

Research Fields

Primary: Agricultural and Trade Econometrics

Secondary: Industrial Organization

Employment

Graduate Student Researcher for Profs. Pierre Mérel and Richard J. Sexton, UC Davis — Fall 2024–Summer 2025

Graduate Student Researcher for Profs. Pierre Mérel and Richard J. Sexton, UC Davis — Summer 2022–Summer 2023

Academic Publications

Mérel, Pierre, **Zhiran Qin**, and Richard J. Sexton. 2023. "Policy-induced expansion of organic farmland: Implications for food prices and welfare." *European Review of Agricultural Economics* 50(4): 1583–1631. [Link]

Consoli, Sarah, Elizabeth A. Fraysse, Natalya Slipchenko, Yi Wang, Jahon Amirebrahimi, **Zhiran Qin**, Neil Yazma, and Travis J. Lybbert. 2022. "A 'Sideways' supply response in California winegrapes." *Journal of Wine Economics* 17(1): 42–63. [Link]

Research Papers

Nutrition Transition and Population Growth: Implications for Land Use and Food Prices **Job Market Paper**

As developing countries grow richer, diets shift from starchy staples toward vegetables, fruits, and animal protein, placing pressure on world agricultural production. Building on Costinot, Donaldson, and Smith (2016), I develop a transparent partial equilibrium model of global crop supply and demand and project country-specific food prices, crop prices, and land use to 2050 under alternative nutrition-transition and population-growth scenarios, informing policy responses to future food demand.

From Trials to Commercialization: Global Market Effects of China's Genetically Modified Soybean and Corn Policies
(*Working Paper*) with Zhiyao Ma and Pierre Mérel

China's move from pilot trials to nationwide commercialization of genetically modified corn and soybeans marks a potentially consequential shift in global crop supply. We develop and calibrate a partial equilibrium model of world crop supply, demand, and trade to quantify how this domestic rollout transmits to international markets. A 20% yield gain on China's soybean acreage raises domestic output substantially but transmits weakly to world prices; welfare gains concentrate in China, while Brazil and the United States bear the largest producer losses.

Contract Prices for Agricultural Products When Buyers Have Heterogeneous Valuations and Binding Capacity Constraints
(*Working Paper*) with Richard J. Sexton

Using duopsony examples and a Hotelling framework, we show that binding capacity constraints severely attenuate competitive bidding in agricultural procurement markets, pushing contract prices toward sellers' outside options even when multiple non-collusive buyers are present. The analysis identifies capacity constraints as an underappreciated channel of buyer market power, with implications for farm prices, farmer welfare, and antitrust policy.

Scholarships, Honors, and Awards

Jastro-Shields Graduate Award 2024, 2025, 2026

Teaching Fields

Microeconomics, Econometrics

Teaching Experience

Teaching Assistant, UC Davis

Microeconomic Analysis (204A/B, graduate)	Fall 2023, Winter 2024
Applied Econometrics (256B, graduate)	Winter 2022
Futures & Options Markets (139)	Spring 2025
Operations Research & Management Science (155)	Winter 2025
Principles of Finance (171)	Spring 2024, Fall 2024

Instructor, UC Davis

Intermediate Microeconomics: Imperfect Competition, Markets & Welfare Economics (100B) Summer 2024

Professional Activities

Conferences

"Policy-induced Expansion of Organic Farmland: Implications for Food Prices and Welfare," 2023 AAEA, Washington, DC (presented by Pierre Mérel) 2023

"Nutrition Transition and Population Growth: Implications for Land Use and Food Prices," 2025 AAEA, Denver, CO 2025

"From Trials to Commercialization: Global Market Effects of China's Genetically Modified Soybean and Corn Policies," 2026 AAEA, Kansas City, MO 2026

"Contract Prices for Agricultural Products When Buyers Have Heterogeneous Valuations and Binding Capacity Constraints," 2026 AAEA, Kansas City, MO 2026

Other

Software: R, Stata, MATLAB, Python, ArcGIS, LaTeX